

Nyaya Sakhi

AI-Powered Dynamic Mental Health Monitoring & Distress Prediction System

MoSJE • NHAA 14566 | SC/ST (PoA) Act, 1989 | Python 3.11 + FastAPI | LangGraph Agentic AI | React + Vite Dashboard | DPDP Act 2023 Compliant | AES-GCM-256 Encryption | 39+ Automated Tests

MedTech / BioTech / HealthTech • Software Category

Nodal Ministry:	Ministry of Social Justice and Empowerment (MoSJE), Government of India
Associated Initiative:	National Helpline Against Atrocities — NHAA 14566
Legal Framework:	Scheduled Castes and Scheduled Tribes (Prevention of Atrocities) Act, 1989

1. Project Overview & Problem Addressed

Victims and witnesses under the **SC/ST (Prevention of Atrocities) Act, 1989**, registered through NHAA 14566 or the Integrated Portal, face prolonged psychological distress *after* complaint registration. Threats, bail of accused, judicial delays, social ostracism, and compensation uncertainty compound their trauma across every stage of the legal process.

THE GAP

Existing government mechanisms focus on legal and financial support. None provide continuous, proactive psychological monitoring of victim well-being throughout the investigation → trial → rehabilitation → compensation cycle.

Nyaya Sakhi fills this gap with a proactive, AI-driven distress prediction system that:

- Continuously interacts with victims across chatbot, IVRS, SMS, WhatsApp, and Web Portal channels
- Fuses voice prosody, text sentiment, behavioral signals, and legal case context into a Dynamic Distress Score
- Predicts deterioration before a crisis emerges and automatically routes interventions
- Provides dashboards at district, state, and national levels for evidence-based oversight

Owning Ministry & Deployment Authority

SINGLE MINISTRY OWNERSHIP

Nyaya Sakhi is designed exclusively for the Ministry of Social Justice and Empowerment (MoSJE) through the National Helpline Against Atrocities (NHAA 14566) and operates under the SC/ST (PoA) Act, 1989. It has NO dependency on or overlap with the Ministry of Women & Child Development, Mission Shakti, or One Stop Centres.

2. Why AI? — The Core Argument

THE ONE-SENTENCE ANSWER

No traditional case-management portal can detect that a victim who sent three short messages after midnight, whose accused was just granted bail, and whose speech pace has slowed by 40% — is at imminent risk of a psychological crisis. Only the longitudinal fusion of language + voice + behaviour + live legal context makes this possible.

What a Normal Portal Does	What Nyaya Sakhi Does Instead
Stores case records	Proactively contacts victim on schedule
Logs victim registration	Measures distress change over time
Tracks FIR / compensation	Detects crisis before victim articulates it
Requires victim to call	Correlates legal events with distress spikes
Reactive — acts after crisis	Auto-dispatches counselor / IVRS call

The AI necessity is established by four signals that **only machine learning can combine at scale**:

Signal	What It Captures	Code Reference
Text Sentiment	Linguistic distress markers: helplessness, fear, grief, suicidal ideation	nlp_agent.py
Voice Prosody	Pitch variance, pause ratio, speaking rate — markers of depression & agitation	speech_agent.py
Behavioural Patterns	Missed check-ins, response latency drift, prolonged silence	behavior_agent.py
Legal Case Context	Bail status, threat reported, hearing postponed, compensation rejected	context_agent.py

3. System Architecture — How It Works

Victim Case Lifecycle Coverage

FIR Filed Initial Registration	Investigation Bail / Threat Flags	Trial Cross-Examination	Rehabilitation Compensation Status	Compensation Claim Tracking
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context_agent.py reads case_stage, compensation_status, accused_bail_status, threat_reported, and hearing_postponed for every victim interaction. A compensation denial adds +0.20 to context risk weight; accused bail adds +0.25; reported threat adds +0.30.

LangGraph Multi-Agent Architecture

Agent	Weight	Input	Key Output
NLP Agent	35-45%	Cleaned message text	distress_score, self_harm_cues, threat_violence_flag
Speech Agent	0-15%	pitch_variance_hz, pause_ratio, speaking_rate_wpm	tone_distress_score, prosodic_cues, tone_label
Behavioral Agent	25%	missed_checkins, response_latency_hours, days_since_last_checkin	behavioral_anomaly_score, disengagement_flag
Context Agent	25-30%	bail_status, threat_reported, case_stage, compensation_status	context_risk_weight, legal_triggers
Fusion Agent	-	All four agent outputs	fused_risk_score [0.0-1.0], risk_tier, explainability_reasons
Escalation Agent	-	fused_risk_score, risk_tier	Alert payload P0-P3, Twilio dispatch, Dashboard push

Fused Score = (NLP * w1) + (Speech * w2) + (Behaviour * w3) + (Context * w4) + Synergy Penalties
w1=0.35-0.45 | w2=0-0.15 | w3=0.25 | w4=0.25-0.30 (speech weight redistributed when no audio)

Risk Classification & Escalation Routing

Tier	Score	Priority	System Action	Alert Reaches
Routine	0.00-0.29	P3	Silent log, next check-in scheduled	Logged only
Watch	0.30-0.49	P3-WATCH	Dashboard badge, monitoring cadence increased	District log
Outreach	0.50-0.74	P2-HIGH	Counselor queue dispatch, follow-up within 24h	District Counselor
Urgent	0.75-1.00	P1-CRITICAL	Auto Twilio Voice + SMS, 30-min acknowledgment	Counselor + District Officer + National Dashboard

A P0-EMERGENCY tier is triggered by the manual SOS panic button (victim-initiated), bypassing NLP scoring with a 5-minute deduplication window.

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4. Multi-Level Dashboard Architecture

District Level	State Nodal Officer	National / Ministry
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Live Triage Roster by risk tier Victim Detail Modal + XAI radar chart Real-time AlertsFeed KPI Cards: active cases, high-risk count One-click manual outreach DPDP Compliance Panel	Cross-district distress metrics Longitudinal trend charts Counselor resource demand forecasting High-risk district heatmap Intervention effectiveness tracking Case-type breakdown by category	Executive summary: victims, critical cases National escalation volume trends Policy analytics: resolution vs distress Searchable national case directory Scheme-wise compensation tracking Evidence base for MoSJE policymakers
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STATE NODAL OFFICER LAYER

Aggregates district-level data, surfaces inter-district comparison, and provides resource allocation recommendations — the intermediary between district counselor dashboards and the national ministry view, ensuring the required three-level hierarchy is operationally complete.

5. Omnichannel Interaction Mechanisms

Channel	Ingress Mechanism	Periodic Outreach?	Voice Analysis?
Telegram	Long-polling bot (@nhaa_14566_sih_bot), voice notes (OGG→WAV)	Yes — Scheduled check-in messages	Yes — Whisper ASR + prosody
WhatsApp	Twilio webhook /webhook/whatsapp-inbound	Yes — Twilio message dispatch	No — Text only
SMS	Twilio fallback when WhatsApp delivery fails	Yes — SMS outreach	No — Text only
IVRS	Twilio Voice IVR, automated outbound call for URGENT tier	Yes — Emergency auto-call	Yes — Prosody via IVR metadata
Web Chat	Floating widget (/api/chat/web), cold-start distress detection	No — Passive (victim-initiated)	No — Text only
Victim Portal	Email OTP auth via Brevo (/api/portal/*)	No — Self-service only	No — Text only

6. Security, Privacy & DPDP Act 2023

Encryption & Storage	Access Control & Consent
• AES-GCM-256 field-level encryption on all victim messages, phone numbers, and PII before (AES-GCM/nonce) via hmac.compare_digest • SQLite with encrypted sensitive fields — no plaintext PII at rest • Automated 730-day data retention policy	• OTP rate-limited to 3 attempts/hour per email • Victim Portal uses allowlist response model — raw risk scores never exposed

7. Explainable AI & Accurate Technical Claims

ACCURATE AI CLAIM — NO MICRO-EXPRESSION ANALYSIS

Nyaya Sakhi does NOT perform video-based facial micro-expression analysis. All acoustic analysis is prosody-based: pitch variance (Hz), pause ratio (%), and speaking rate (WPM) — extracted from voice note audio only. This is speech-acoustic analysis, not facial recognition.

What the System Actually Measures

Claimed Capability	Accurate Technical Description	Code Reference
Emotion AI	28-class emotion classification on text using fine-tuned IndicBERTv2	nlp_agent.py:58
Voice Stress Analysis	Acoustic prosody: pitch_variance <10Hz → flat/withdrawn; pause_ratio >0.35 → psychomotor slowing; pitch >55Hz + energy >0.12 → agitation	speech_agent.py:28–47
Sentiment Analysis	Zero-shot MNLI + GoEmotions multilabel scoring; regex crisis-pattern fallback when inference API offline	nlp_agent.py:25–44
Predictive Risk	Behavioral velocity: missed check-ins $\geq 3 \rightarrow +0.50$; latency ratio $\geq 3 \times$ baseline $\rightarrow +0.30$; 7+ days silence $\rightarrow +0.35$	behavior_agent.py:18–40
XAI / Explainability	Every fusion output carries explainability_reasons list — per-agent sub-scores, legal triggers, prosody cues rendered as radar chart	fusion_agent.py:35–76
Synergy Detection	Synergy multiplier ($\times 1.35$) applied when bail_granted + disengagement_flag + nlp_score ≥ 0.4 occur simultaneously	fusion_agent.py:73–75

Dynamic Distress Score — Correct Framing

HOW TO DESCRIBE THIS TO A JUDGE

'The score is an AI-derived risk indicator for triage prioritisation — not a clinical diagnosis, not a DSM-5 severity rating, and not a medically validated psychological measurement. It tells a counsellor who needs attention now compared to others. All decisions remain with the human counsellor.'

Thresholds configurable in backend/config.py: URGENT ≥ 0.75 , OUTREACH ≥ 0.50 , WATCH ≥ 0.30 . Every escalation alert contains human_in_the_loop_status: 'Awaiting Counselor Review' until a counsellor explicitly acknowledges.

8. Multilingual & Inclusivity

- IndicBERTv2 natively supports 24 Indian languages via multilingual pre-training — fine-tuned model (Robbiinn/nyaya-sakhi-crisis-indicbert) inherits this cross-lingual capability.
- Primary conversational support: Hindi and English via Groq LLM multilingual system prompts.
- IVRS outreach scripts are configurable per language; Twilio supports regional TTS voices.
- Fine-tuning datasets: DASS-21 distress corpus, GoEmotions, and Dreddit — cleaned for atrocity-context patterns.
- Architecture supports plug-in language adapters for regional expansion (Tamil, Telugu, Bengali, Marathi, Odia).

9. Judge Q&A — Anticipated Questions & Accurate Answers

1

Which ministry owns this system, and why is it MoSJE and not the Ministry of Women & Child Development?

Built exclusively for the Ministry of Social Justice and Empowerment (MoSJE) under the SC/ST (Prevention of Atrocities) Act, 1989. Beneficiaries are victims registered through NHAA 14566 — the national helpline managed by MoSJE specifically for SC/ST atrocity cases.

The Ministry of Women & Child Development operates Mission Shakti and One Stop Centres — a separate mandate. There is no functional overlap. Nyaya Sakhi does not reference or deploy through WCD infrastructure.

2

What exactly does your AI analyse in voice messages? Do you do facial expression analysis?

No facial expression analysis. The system analyses acoustic prosody from voice recordings:

- Pitch variance (Hz): Below 10 Hz = flat, withdrawn affect; above 55 Hz + high energy = agitation or panic.
- Pause ratio (%): Above 35% = psychomotor slowing associated with severe distress.
- Speaking rate (WPM): Below 90 WPM alongside high pause ratio = psychomotor slowing.

Extracted from Whisper ASR followed by Wav2Vec2 acoustic feature analysis — audio only, not video. (speech_agent.py:28–47)

3

Is your 0–100% Distress Score clinically validated? Can it be used as a psychiatric diagnosis?

No, and we are explicit about this. The Dynamic Distress Score is an AI-derived triage indicator — not a clinical diagnosis. It answers: 'Who needs a counsellor's attention most urgently right now?'

Not a DSM-5 severity rating or PHQ-9 equivalent. Every alert includes mandatory human_in_the_loop_status: 'Awaiting Counselor Review'. The AI surfaces priority; the licensed counsellor makes all clinical decisions.

4

Why does this need AI? Why can't a normal case management portal do the same thing?

A traditional portal stores and displays case data. It cannot:

- Detect change over time — Behavioral Agent tracks response latency drift (usually 2h, now 24h) and consecutive missed check-ins.
- Interpret language — NLP on 24 Indian languages to detect hopelessness or suicidal ideation from natural conversation text.
- Read vocal signals — pitch flatness and speaking rate slowdown are invisible to a case manager reviewing a text record.
- Synthesise all signals simultaneously — the synergy multiplier ($\times 1.35$ when bail granted + disengagement + high NLP distress co-occur).
- Act proactively — automatically dispatch a Twilio IVR emergency call without waiting for the victim to reach out.

5

How does your system handle a victim who says 'I am fine' but is actually in distress?

Three complementary safeguards:

1. Sentiment Guardrail (nlp_agent.py:29–45): Suppresses false positives for genuinely positive text, but does NOT suppress when crisis-pattern words co-occur. 'I am fine but the accused threatened me' still triggers threat_violence_flag = True.
2. Voice Prosody Override: Flat pitch (<10 Hz) or prolonged pauses (>35%) in the voice note contribute a speech distress score independently of the text.
3. Behavioral Context: 3 missed check-ins + message at 3 AM = high anomaly score regardless of text content.

6 Does your system support Hindi and regional languages?

Yes. The NLP model is fine-tuned IndicBERTv2 (Robbiinn/nyaya-sakhi-crisis-indicbert), which natively supports 24 Indian languages through multilingual pre-training. The model produces distress scores directly from Hindi, Bengali, Tamil, and other regional language text.

The Groq LLM counselor companion uses multilingual system prompts. IVRS scripts support regional TTS delivery via Twilio.

7 What happens when the AI model inference service is offline?

Explicit offline fallback in `nlp_agent.py:109–138`. When the IndicBERT API is unreachable:

- Falls back to regex-based crisis pattern scanner (kill, suicide, threat, rape, danger, harm, attack).
- If crisis patterns detected: `distress_score` = 0.95, all critical flags set.
- If no crisis patterns: `distress_score` = 0.05 and `analysis_source` = 'indicbertv2_offline_standby' recorded.

The system always provides a triage signal, even without the ML model.

8 How do you protect victim privacy? What if data is leaked?

- AES-GCM-256 encryption on all sensitive fields before writing to the database (`privacy.py`).
- Consent architecture: every channel records `consent_flag` and `consent_timestamp`.
- PII masking in NLP pipeline: `clean_text()` masks phone numbers [PHONE] and emails [EMAIL] before sending to the ML model.
- Officer-gated purge: `POST /api/officer/purge` uses `hmac.compare_digest` constant-time comparison.
- Victim Portal allowlist: raw distress scores are never exposed to the victim-facing client.
- 730-day automated retention: compliant with DPDP Act 2023.

9 What if the AI gives a wrong prediction?

- Sentiment Guardrail: Genuine positive messages suppress false distress scoring before reaching the counsellor queue.
- XAI Explanation: Every alert includes a human-readable `clinical_reasons` list — counsellor sees exactly why the score was elevated and can rapidly dismiss if reasons are not compelling.
- Risk Tiers as Filter: Only OUTREACH (≥ 0.50) and URGENT (≥ 0.75) trigger active counsellor notification. ROUTINE and WATCH are passive only.
- Human-in-the-Loop Mandatory: The system cannot take clinical action autonomously.

10 Does your system cover victims through the entire judicial process?

Yes. `context_agent.py` reads these lifecycle fields for every victim interaction:

- `case_stage`: Trial / Cross-Examination stages add +0.20 to context risk weight.
- `accused_bail_status`: 'Granted' adds +0.45 — the highest safety vulnerability penalty.
- `threat_reported`: Adds +0.50 when active threat is logged.
- `hearing_postponed`: Institutional delay adds +0.15.
- `compensation_status`: 'Rejected' adds +0.20 — models financial and psychological distress of a denied claim.

All five fields are first-class columns in the victims database table.

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What is the difference between Track 1 (Groq LLM) and Track 2 (LangGraph)?

- Track 1 — Groq LLM: Victim-facing conversational response — warm, empathetic, multilingual text in ~400ms. Provides emotional support, surfaces NHAA 14566 helpline. Does NOT make risk assessments.
 - Track 2 — LangGraph Multi-Agent Pipeline: Runs silently in the background. 4 specialist agents analyse the message, outputs are fused into a risk score, XAI explanation generated, escalation agent decides on alerts.
- The victim always receives a human-feeling response (Track 1) while the system simultaneously makes a risk decision (Track 2) without introducing clinical language into the victim-facing interaction.

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How do you ensure victims are not re-traumatised by the chatbot?

The Groq LLM counselor persona (counselor_persona.py) operates under hard-coded non-clinical guardrails:

- Never identifies itself as a doctor, psychologist, or clinical professional.
- Never makes predictions about court outcomes, bail, or compensation decisions.
- Always surfaces NHAA 14566 helpline and emergency number 112 in any distress context.
- Responses are warm, validating, and non-alarmist.
- 39+ automated tests in test_counselor_persona.py verify helpline inclusion, no false medical claims.

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What are the genuine innovation components?

1. Multi-signal temporal fusion: No existing government victim support system fuses text sentiment + voice prosody + engagement velocity + live legal case context into a single calibrated risk score per interaction.
2. Synergy Multiplier: Detection of concurrent risk co-occurrence (bail granted + disengagement + high NLP distress simultaneously) — not present in standalone NLP or voice systems.
3. Proactive omnichannel check-ins: The system contacts victims — they do not need to initiate. Most AI helplines are purely reactive.
4. Longitudinal distress tracking: Score evolves over time; a rising trend is detected even if each individual score is moderate.
5. XAI-first architecture: Explainability built in from day one — every alert includes a clinical_reasons list built from each agent's contribution.

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Can you demo this system live?

Yes. One command: `python backend/run_project.py` — starts FastAPI backend, React dashboard, and Telegram bot.

Demo scenarios: send a distress message to @nhaa_14566_sih_bot → see triage result with XAI radar chart; send a voice note → see pitch variance, pause ratio, and tone label; press SOS panic button → observe P0-EMERGENCY alert and Twilio voice call dispatch; use the Live Channel Simulator for any channel without a live phone.

Full test suite: `python -m pytest tests/ -q` — 39+ tests covering compliance, guardrails, escalation thresholds, and end-to-end flow.

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What is the SOS panic button and how does it differ from AI-detected escalation?

- AI-Detected Escalation: LangGraph pipeline infers risk from language, voice, and behaviour — runs on every inbound message automatically.
- Manual SOS Panic Button: Victim-initiated from web portal or Telegram. Skips all NLP scoring — goes directly to P0-EMERGENCY priority. Triggers Twilio Voice calls to pre-configured SOS recipients. 5-minute deduplication window prevents accidental double-press. Logged as manual_sos trigger type.

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How does the system handle a victim who stops responding entirely?

Handled by the Behavioral Agent (behavior_agent.py):

- 7+ days since last check-in: Adds +0.35 to behavioral anomaly score, sets disengagement_flag = True.
- 3+ consecutive missed check-ins: Adds +0.50 — alone can push the score into the OUTREACH tier.
- Synergy with legal context: 7+ days silence + bail just granted → synergy multiplier (×1.35) → likely URGENT tier → automated IVRS call triggered.

The most at-risk victims are often the least likely to call. Nyaya Sakhi detects the ABSENCE of engagement as a distress signal.

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What interventions does the system recommend and execute?

Context-aware recommendations: high fear/grief + bail granted → Counselling + witness protection alert; voice tremor + night messaging → Urgent counsellor callback + medical referral; compensation 'Rejected' → Legal aid + financial assistance scheme; suicidal ideation → Immediate IVRS emergency call + 24x7 counsellor escalation.

The system automatically executes the communication layer (Twilio IVR + SMS) for OUTREACH and URGENT tiers. Clinical intervention decisions (scheduling counselling, protection applications) remain with the human counsellor.

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How is the distress score explained to the counsellor?

Every alert carries:

- clinical_reasons: Plain-language strings — '■ Explicit self-harm markers detected', '■■ Accused Released on Bail', '■■■ Vocal Tremor / Acute Agitation', '■ 4 consecutive missed check-ins', '■ Synergistic Risk Escalation detected.'
- weights_used: The actual weights applied in the fusion for this specific interaction.
- Radar Chart in Dashboard: The Victim Detail Modal renders a multi-modal radar chart showing NLP, Speech, Behavioral, and Context sub-scores visually.

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This is a SQLite prototype — how does it scale to national deployment?

- Database layer abstracted through database.py — parameterised SQL; migration to PostgreSQL is a connection string change.
- FastAPI backend is stateless and horizontally scalable behind a load balancer.
- LangGraph agents are isolated functional units — containerisable as separate microservices.
- render.yaml cloud deployment blueprint is included in the repository.
- National deployment: managed PostgreSQL (AWS RDS / NIC cloud), GPU-hosted ML inference, Government-approved IVRS provider.

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What is the single most important thing Nyaya Sakhi does?

Nyaya Sakhi detects that a victim who has gone silent for a week, whose accused was just granted bail, and whose last voice message showed flat pitch and slow speech — is at imminent crisis risk. It then automatically calls them. No existing government system does this without the victim initiating contact.

Everything else — multilingual NLP, XAI radar charts, DPDP compliance, dashboards — supports this one capability: proactive, early detection of psychological deterioration before a crisis, across hundreds of SC/ST atrocity victims simultaneously, with automatic intervention routing.

